

AI Agents & Automation — Professional Masterclass

Design reliable agentic workflows and automate real business processes.

This is an original, project-first professional course manual created for educational and commercial course-development use.

The material is designed to be taught, adapted, branded and expanded. It emphasizes practical engineering, portfolio projects, assessments and production thinking rather than certificate collection. Technology names are used for educational context; specific provider pricing, product limits and interfaces should be verified before publication because they change over time.

Course Architecture

- 7 large modules with lessons, practical exercises, assessments and projects.
- Recommended delivery: 10–14 weeks at 6–10 hours per week, or self-paced.
- Primary outcome: a portfolio-ready capstone plus reusable professional skills.
- Suggested teaching method: concept → demonstration → guided lab → independent project → assessment.

Module Roadmap

1	Automation Foundations
2	Agent Architecture and Tool Use
3	n8n-Style Workflow Automation and Integrations
4	MCP, Tools and Connected AI Systems
5	Business Process Agents
6	Reliability, Security and Economics
7	Capstone: AI Automation Agency / SaaS

Module 1 — Automation Foundations

Learning objectives

- Automation vs AI automation
- Workflow mapping
- APIs, webhooks and event-driven design
- Triggers, actions and conditions
- Idempotency and retries
- Human-in-the-loop patterns
- Mini-project: automated lead intake

Core lesson

AI automation begins with process design, not with an agent framework. Map the current workflow: trigger, inputs, decisions, actions, outputs and exceptions. Identify which steps are deterministic and which require judgment. Deterministic steps belong in normal workflow code; AI should be introduced where language, classification, extraction or flexible reasoning adds value.

APIs and webhooks connect systems. A webhook receives an event, validates its signature when applicable, records an event identifier and triggers processing. Idempotency prevents the same event from creating duplicate records. Every automation should have a clear retry policy and a failure destination.

Conditions can be simple rules or model-based decisions. Keep high-impact decisions explainable and reviewable. For example, an AI classifier can prioritize a support request, but a deterministic policy can decide whether the ticket may be closed.

Human-in-the-loop design is a strength, not a weakness. Let automation prepare a draft, recommendation or proposed action while a person approves sensitive steps. Gradually increase automation only when evaluation demonstrates safety and reliability.

Practical lab

- Map a manual business process into a workflow diagram.
- Build a webhook receiver with event deduplication.
- Create a classifier that routes requests to three queues.
- Add a human approval step before an external action.
- Document failure and retry behavior.

Assessment

Assessment: Choose five steps from a business process and justify whether each should be deterministic, AI-assisted or human-controlled.

Module project

Project: Lead Intake Automation — receive a form/webhook, extract lead details, score the lead, route it, notify a team and record the audit trail.

Module 2 — Agent Architecture and Tool Use

Learning objectives

- Agent loop fundamentals
- Tool definitions
- State and memory
- Planning vs execution
- Bounded autonomy
- Multi-agent patterns
- Agent observability

Core lesson

An agent is a software system in which a model helps decide the next action using a set of available tools and context. The important engineering work is the boundary around the model. Define what it can see, what it can do, how many steps it can take and when it must stop.

Use typed tool definitions. A CRM search tool should accept a query and return structured records; it should not expose arbitrary database execution. Separate read tools from write tools and apply stronger controls to writes.

State includes the task, prior observations, tool results and progress. Long-running agents benefit from checkpoints. Memory should be scoped: session memory for the current task, user memory for durable preferences, and organizational memory for shared knowledge.

Multi-agent systems can divide work among specialist roles, but they add communication overhead and failure modes. Start with a single bounded agent. Add multiple agents only when a clear specialization improves evaluation results.

Observability should show the trace: input → decision → tool call → result → next decision → final output. This makes debugging possible and provides evidence for improving prompts and tools.

Practical lab

- Build a single agent with three tools.
- Add a step budget and timeout.
- Persist checkpoints in a database.
- Compare single-agent and two-agent designs on one task.
- Create a trace viewer or structured log.

Assessment

Assessment: Explain why adding agents can reduce reliability. Design a bounded architecture with a maximum cost and execution time.

Module project

Project: Operations Agent — search internal records, summarize findings, propose actions and require approval before writing changes.

Module 3 — n8n-Style Workflow Automation and Integrations

Learning objectives

- Workflow builders
- Nodes and credentials
- HTTP/API integration
- Database automation
- Email and messaging workflows
- Scheduled jobs
- Error workflows and replay

Core lesson

Visual workflow platforms such as n8n are useful because they make integrations fast to build and easy to inspect. Learn the concepts even if you later implement workflows in code. A workflow is a directed graph of triggers, transformations, conditions and actions.

Credentials should be stored in the platform's secret system or a dedicated secret manager. Never place API keys in ordinary workflow text. Normalize incoming data early and validate required fields.

HTTP integrations require careful handling of status codes, pagination, timeouts and rate limits. Database workflows should use parameterized queries and transactions where consistency matters.

Scheduled automation is useful for reports, synchronization and maintenance. Design jobs so they can be safely rerun. For every workflow, create an error path that captures the event, error message, execution identifier and enough context to reproduce the problem.

A mature automation platform also needs replay. If a downstream service was unavailable, an operator should be able to retry the failed item without duplicating successful side effects.

Practical lab

- Build an API-to-database workflow.
- Connect a form submission to email and a database.
- Create a scheduled daily summary workflow.
- Add an error branch and persistent execution log.
- Test replay without creating duplicates.

Assessment

Assessment: Design a workflow that remains safe if a webhook is delivered three times or an API fails halfway through.

Module project

Project: Business Automation Hub — intake, validation, enrichment, routing, notification, database storage and error recovery.

Module 4 — MCP, Tools and Connected AI Systems

Learning objectives

- MCP concepts
- Resources and tools
- Tool discovery
- Authorization boundaries
- Context selection
- Building an MCP-style server
- Testing connected assistants

Core lesson

Connected AI systems need standardized interfaces for tools and resources. MCP is one important interoperability pattern. The core idea is to expose capabilities with explicit schemas and controlled context so compatible AI clients can use them.

Treat every connected capability as a security boundary. Authentication identifies the caller; authorization decides what that caller may access. A tool description must never be mistaken for authorization. The server must enforce permissions.

Resource design also matters. Instead of dumping an entire database into context, expose targeted resources or search operations. The goal is to provide the minimum useful context for a task.

When building an MCP-style server, begin with read-only tools. Add write operations after tests and approval flows are in place. Validate inputs, sanitize outputs and log calls.

Testing should include valid calls, malformed arguments, unauthorized access, large results, timeouts and malicious instructions embedded in resources. Interoperability is valuable only when the connected system remains trustworthy.

Practical lab

- Design a tool/resource catalog for an internal app.
- Implement three read-only capabilities.
- Add authorization to every capability.
- Test malicious and malformed inputs.
- Document tool schemas for another developer.

Assessment

Assessment: Explain the difference between tool discoverability and permission to use a tool.

Module project

Project: Connected Workspace Server — expose safe search, record lookup and report-generation tools to an AI client.

Module 5 — Business Process Agents

Learning objectives

- Sales agents
- Customer support agents
- Research agents
- Document-processing agents
- Recruiting/admin workflows
- Personal productivity agents
- Choosing the right automation

Core lesson

Business agents should be designed around measurable outcomes. A sales agent might research a prospect and draft an outreach message. A support agent might classify a ticket and retrieve a solution. A research agent might gather evidence and produce a report. Each workflow has different risk and evaluation requirements.

Sales automation must avoid spam and respect communication policies. Support agents need strong grounding and escalation when evidence is insufficient. Research agents need source tracking and explicit uncertainty. Document agents need extraction validation because one wrong field can propagate into downstream systems.

Recruiting and administrative workflows require additional care because sensitive personal information may be involved. Limit data access, define retention rules and avoid automated decisions where human review is necessary.

Productivity agents are often the safest place to start: summarize meetings, organize notes, draft documents or transform data. They can demonstrate value while keeping side effects under user control.

Select automation candidates using a simple score: $\text{frequency} \times \text{time saved} \times \text{value per successful execution} \div \text{risk and implementation complexity}$. Automate repetitive, well-bounded processes before ambiguous, high-impact ones.

Practical lab

- Choose one process using the value/risk score.
- Build a support triage workflow.
- Build a document extraction workflow with validation.
- Create an evidence-based research agent.
- Compare time saved before and after automation.

Assessment

Assessment: Rank six automation ideas and defend the order using value, risk and complexity.

Module project

Project: Business Agent Suite — implement two complete agents for different workflows and compare their reliability and economics.

Module 6 — Reliability, Security and Economics

Learning objectives

- Prompt injection
- Data access controls
- Secrets and credentials
- Rate limiting
- Cost budgets
- Evaluation and monitoring
- Incident response

Core lesson

Automation magnifies both value and mistakes. A workflow that runs once per day can create thousands of incorrect actions if a bug remains undetected. Reliability controls therefore belong in the architecture from the beginning.

Prompt injection occurs when untrusted content attempts to influence an AI system's instructions. Treat emails, webpages, documents and retrieved text as data, not trusted policy. Keep permissions in code and never allow the model to bypass authorization.

Set budgets. Limit model calls per workflow, tool calls per task and total execution time. Track cost by workflow and customer. Use smaller models for simple extraction and classification when quality remains acceptable.

Evaluation should include normal cases, edge cases and adversarial cases. For automation, measure not only answer quality but action correctness: did the workflow create the correct record, send the right message and avoid duplicates?

Incident response should include stopping automation, identifying affected executions, repairing data and documenting the root cause. Add kill switches for important workflows.

Practical lab

- Threat-model one automation.
- Implement a global execution budget.
- Add a kill switch to a workflow.
- Build an evaluation set with adversarial inputs.
- Calculate cost per successful automated task.

Assessment

Assessment: Design controls for a workflow that can send external messages and modify customer records.

Module project

Project: Automation Control Center — monitoring, budgets, audit logs, failure handling and a workflow kill switch.

Module 7 — Capstone: AI Automation Agency / SaaS

Learning objectives

- Offer design
- Client discovery
- Reusable workflow templates
- Multi-tenant architecture
- Pricing and margins
- Deployment and maintenance
- Portfolio and sales strategy

Core lesson

The final goal is to turn automation skills into a repeatable product or service. Start with a niche where the same workflow appears across many customers. Examples include lead qualification, customer support triage, document intake, reporting and internal knowledge search.

Create a standardized implementation package. Include discovery questions, process map, data requirements, integration checklist, test cases, deployment procedure and monitoring plan. Reusable templates reduce delivery time and increase margin.

For SaaS, isolate tenants at every layer: database queries, files, retrieval indexes, logs and credentials. For services, establish clear boundaries around what the automation will and will not do.

Price around value while protecting your margin. Estimate setup time, recurring infrastructure, model usage, maintenance and support. A fixed setup fee plus recurring management can be easier to understand than billing every API call.

Your portfolio should show before/after process metrics: minutes saved, error rate, response time or throughput. A working demo and measured result are stronger than a list of tools.

Practical lab

- Pick one niche and one repeatable workflow.
- Build a reusable template.
- Create a client discovery checklist.
- Calculate implementation and monthly support costs.
- Publish a case study with measured results.

Assessment

Assessment: Present an automation offer with architecture, security controls, pricing, ROI and maintenance plan.

Module project

Capstone: AI Automation Business — deliver a production workflow for a realistic customer, with monitoring, audit logs, documentation and a repeatable deployment package.

Instructor & Commercial Packaging Guide

- Turn each module into 3–6 video lessons plus one guided lab.
- Provide a downloadable starter repository and a separate completed reference implementation.
- Add quizzes after each lesson and a practical assessment after each module.
- Keep project requirements stable while allowing learners to choose their own domain.
- For commercial delivery, add your own branding, instructor examples, screenshots, exercises and community/support layer.
- Avoid promising jobs, income or certification unless you actually provide the relevant accreditation or service.

Suggested Course Completion Certificate Criteria

- Complete at least 5 of 7 module projects.
- Pass the capstone assessment.
- Submit a working portfolio artifact and technical documentation.
- Demonstrate the ability to explain design decisions and limitations.

Final Portfolio Checklist

- Public or private demo URL where appropriate.
- Architecture diagram.
- README with setup and usage.
- Test/evaluation evidence.
- Security and limitations section.
- Short case study describing the problem and measurable result.